

Import Library

```
In [ ]: # Importing the pandas library for data manipulation and analysis
import pandas as pd

# Importing NumPy for numerical operations and handling arrays
import numpy as np

# Importing Matplotlib's pyplot module for data visualization
import matplotlib.pyplot as plt
```

Import data using read_csv()

```
In [ ]: # 'encoding="latin1"' is used to properly read special characters (Like accents
company=pd.read_csv('/content/Unicorn_Companies.csv', encoding='latin1')

In [ ]: # Display the contents of the 'data' variable (usually a DataFrame)

company
```

Out[]:

	Company	Valuation (\$B)	Date Joined	Country	City	Industry	Select Inven
0	Bytedance	\$140	4/7/2017	China	Beijing	Artificial intelligence	Sequoia C China, SIC Investmen
1	SpaceX	\$100.3	12/1/2012	United States	Hawthorne	Other	Founders Draper I Jurve Rot
2	Stripe	\$95	1/23/2014	United States	San Francisco	Fintech	Khosla Ven LowercaseCa cap
3	Klarna	\$45.6	12/12/2011	Sweden	Stockholm	Fintech	Institut Venture Par Sequoia Ca
4	Epic Games	\$42	10/26/2018	United States	Cary	Other	Te Holdings, Smash Ver
...	...	...	...	...	...	...	...
1032	Timescale	\$1	2/22/2022	United States	New York	Internet software & services	New Ente Assoc Benchmark S
1033	Scalapay	\$1	2/23/2022	Italy	Milan	Fintech	Fasanara Ca Tiger C Manage
1034	Omada Health	\$1	2/23/2022	United States	San Francisco	Health	U.S. Ve Partner: Ca Andreess
1035	BlueVoyant	\$1	2/23/2022	United States	New York	Cybersecurity	8VC, Li Strategic Ca Eden Globa
1036	Veev	\$1	2/24/2022	United States	San Mateo	Internet software & services	Zeev Ven Bond, Fift Ver

1037 rows × 13 columns



```
In [ ]: # Display the first 2 rows of the DataFrame
company.head(2)
```

Out[]:

	Company	Valuation (\$B)	Date Joined	Country	City	Industry	Select Investors	Founders
0	Bytedance	\$140	4/7/2017	China	Beijing	Artificial intelligence	Sequoia Capital China, SIG Asia Investments, S...	20
1	SpaceX	\$100.3	12/1/2012	United States	Hawthorne	Other	Founders Fund, Draper Fisher Jurvetson, Rothen...	20



The GroupBy() function

```
In [ ]: # Grouping the DataFrame by the 'Industry' column.
company.groupby('Industry')
```

Out[]: <pandas.core.groupby.generic.DataFrameGroupBy object at 0x79f13e9c07d0>

```
In [ ]: # Grouping the DataFrame by the 'Industry' column and storing the result in the
sectore=company.groupby('Industry')
```

```
In [ ]: # Displaying the GroupBy object stored in 'sectore'
# This will not show actual grouped data until an aggregation function is applied
sectore
```

Out[]: <pandas.core.groupby.generic.DataFrameGroupBy object at 0x79f13c088f50>

```
In [ ]: # Getting the total number of rows (companies) in the DataFrame
len(company)
```

Out[]: 1037

```
In [ ]: # Getting the total number of unique industries in the 'Industry' column
len(sectore)
```

Out[]: 33

```
In [ ]: # Getting the number of companies in each industry
sectore.size()
```

Out[]:

0

Industry	
500 Global, Rakuten Ventures, Golden Gate Ventures	1
Andreessen Horowitz, DST Global, IDG Capital	1
Artificial Intelligence	7
Artificial intelligence	71
Auto & transportation	29
B Capital Group, Monk's Hill Ventures, Dynamic Parcel Distribution	1
Consumer & retail	25
Cybersecurity	49
Data management & analytics	41
Dragonfly Captial, Qiming Venture Partners, DST Global	1
E-commerce & direct-to-consumer	107
Edtech	28
Fintech	205
Finttech	1
Hardware	33
Health	69
Hopu Investment Management, Boyu Capital, DC Thomson Ventures	1
Internet software & services	192
Jungle Ventures, Accel, Venture Highway	1
Kuang-Chi	1
Mobile & telecommunications	37
Mundi Ventures, Doqing Capital Partners, Activant Capital	1
Other	56
Sequoia Capital China, ING, Alibaba Entrepreneurs Fund	1
Sequoia Capital China, Shunwei Capital Partners, Qualgro	1
Sequoia Capital, Thoma Bravo, Softbank	1
SingTel Innov8, Alpha JWC Ventures, Golden Gate Ventures	1
Supply chain, logistics, & delivery	57
Temasek, Guggenheim Investments, Qatar Investment Authority	1
Tiger Global Management, Tiger Brokers, DCM Ventures	1
Travel	14
Vertex Ventures SE Asia, Global Founders Capital, Visa Ventures	1

0

Industry

Vision Plus Capital, GSR Ventures, ZhenFund	1
---	---

dtype: int64

```
In [ ]: # Sorting the industries by the number of companies in descending order  
sectore.size().sort_values(ascending=False)
```

Out[]:

0

Industry	
Fintech	205
Internet software & services	192
E-commerce & direct-to-consumer	107
Artificial intelligence	71
Health	69
Supply chain, logistics, & delivery	57
Other	56
Cybersecurity	49
Data management & analytics	41
Mobile & telecommunications	37
Hardware	33
Auto & transportation	29
Edtech	28
Consumer & retail	25
Travel	14
Artificial Intelligence	7
500 Global, Rakuten Ventures, Golden Gate Ventures	1
Hopu Investment Management, Boyu Capital, DC Thomson Ventures	1
Finttech	1
Dragonfly Capital, Qiming Venture Partners, DST Global	1
B Capital Group, Monk's Hill Ventures, Dynamic Parcel Distribution	1
Andreessen Horowitz, DST Global, IDG Capital	1
Mundi Ventures, Doqing Capital Partners, Activant Capital	1
Kuang-Chi	1
Jungle Ventures, Accel, Venture Highway	1
Sequoia Capital China, Shunwei Capital Partners, Qualgro	1
Sequoia Capital China, ING, Alibaba Entrepreneurs Fund	1
SingTel Innov8, Alpha JWC Ventures, Golden Gate Ventures	1
Sequoia Capital, Thoma Bravo, Softbank	1
Temasek, Guggenheim Investments, Qatar Investment Authority	1
Tiger Global Management, Tiger Brokers, DCM Ventures	1
Vertex Ventures SE Asia, Global Founders Capital, Visa Ventures	1

0

Industry

Vision Plus Capital, GSR Ventures, ZhenFund	1
---	---

dtype: int64

```
In [ ]: # Displaying the first company entry from each industry group  
sectore.first()
```

Out[]:

	Company	Valuation (\$B)	Date Joined	Country	City
Industry					
500 Global, Rakuten Ventures, Golden Gate Ventures	Carousell	\$1.1	9/15/2021	Singapore	E-commerce & direct-to-consumer
	Cider	\$1	9/2/2021	Hong Kong	E-commerce & direct-to-consumer
Artificial Intelligence	Uniphore	\$2.5	2/16/2022	United States	Palo Alto
Artificial intelligence	Bytedance	\$140	4/7/2017	China	Beijing
Auto & transportation	ZongMu Technology	\$11.4	6/3/2021	China	Shanghai
B Capital Group, Monk's Hill Ventures, Dynamic Parcel Distribution	Ninja Van	\$1	9/27/2021	Singapore	Supply chain, logistics, & delivery
Consumer & retail	Yuanqi Senlin	\$15	3/1/2020	China	Beijing
Cybersecurity	Tanium	\$9	3/31/2015	United States	Kirkland
Data management & analytics	Databricks	\$38	2/5/2019	United States	San Francisco
Dragonfly Captial, Qiming Venture Partners, DST Global	Matrixport	\$1.05	6/1/2021	Singapore	Fintech
E-commerce & direct-to-consumer	Fanatics	\$27	6/6/2012	United States	Jacksonville
Edtech	BYJU's	\$21	7/25/2017	India	Bengaluru

	Company	Valuation (\$B)	Date Joined	Country	City	
Industry						
Fintech	Stripe	\$95	1/23/2014	United States	San Francisco	1
Finttech	Digital Currency Group	\$10	11/1/2021	United States	New York	
Hardware	DJI Innovations	\$15	1/23/2015	China	Shenzhen	
Health	Devoted Health	\$12.6	10/16/2018	United States	Waltham	
Hopu Investment Management, Boyu Capital, DC Thomson Ventures	Trax	\$1.3	7/22/2019	Singapore	Artificial intelligence	
Internet software & services	Canva	\$40	1/8/2018	Australia	Surry Hills	
Jungle Ventures, Accel, Venture Highway	Moglix	\$2.6	5/17/2021	Singapore	E-commerce & direct-to-consumer	
Kuang-Chi	HyalRoute	\$3.5	5/26/2020	Singapore	Mobile & telecommunications	
Mobile & telecommunications	Niantic	\$9	11/24/2017	United States	San Francisco	
Mundi Ventures, Doqing Capital Partners, Activant Capital	bolttech	\$1	7/1/2021	Singapore	Fintech	
Other	SpaceX	\$100.3	12/1/2012	United States	Hawthorne	
Sequoia Capital China, ING, Alibaba Entrepreneurs Fund	WeLab	\$1	11/8/2017	Hong Kong	Fintech	
Sequoia Capital China, Shunwei Capital Partners, Qualgro	PatSnap	\$1	3/16/2021	Singapore	Internet software & services	

	Company	Valuation (\$B)	Date Joined	Country	City
Industry					
Sequoia Capital, Thoma Bravo, Softbank	FTX	\$32	7/20/2021	Bahamas	Fintech
SingTel Innov8, Alpha JWC Ventures, Golden Gate Ventures	Carro	\$1	6/14/2021	Singapore	E-commerce & direct-to-consumer
Supply chain, logistics, & delivery	Instacart	\$39	12/30/2014	United States	San Francisco
Temasek, Guggenheim Investments, Qatar Investment Authority	ONE	\$1	12/8/2021	Singapore	Internet software & services
Tiger Global Management, Tiger Brokers, DCM Ventures	Amber Group	\$3	6/21/2021	Hong Kong	Fintech
Travel	OYO Rooms	\$9.6	9/25/2018	India	Gurugram
Vertex Ventures SE Asia, Global Founders Capital, Visa Ventures	NIUM	\$1	7/13/2021	Singapore	Fintech
Vision Plus Capital, GSR Ventures, ZhenFund	Advance Intelligence Group	\$2	9/23/2021	Singapore	Artificial intelligence

```
In [ ]: # Displaying the last company entry from each industry group
        sectore.last()
```

Out[]:

	Company	Valuation (\$B)	Date Joined	Country	City
Industry					
500 Global, Rakuten Ventures, Golden Gate Ventures	Carousell	\$1.1	9/15/2021	Singapore	E-commerce & direct-to-consumer
	Cider	\$1	9/2/2021	Hong Kong	E-commerce & direct-to-consumer
Artificial Intelligence	Haomao.AI	\$1	12/22/2021	China	Beijing
Artificial intelligence	Betterfly	\$1	2/1/2022	Chile	Santiago
Auto & transportation	Socar	\$1	10/16/2020	South Korea	Jeju-do
B Capital Group, Monk's Hill Ventures, Dynamic Parcel Distribution	Ninja Van	\$1	9/27/2021	Singapore	Supply chain, logistics, & delivery
Consumer & retail	Kopi Kenangan	\$1	12/27/2021	Indonesia	Jakarta
Cybersecurity	BlueVoyant	\$1	2/23/2022	United States	New York
Data management & analytics	Dune Analytics	\$1	2/2/2022	Norway	Oslo
Dragonfly Capital, Qiming Venture Partners, DST Global	Matrixport	\$1.05	6/1/2021	Singapore	Fintech
E-commerce & direct-to-consumer	SellerX	\$1	12/9/2021	Germany	Berlin

	Company	Valuation (\$B)	Date Joined	Country	City
Industry					
Edtech	Vedantu	\$1	9/29/2021	India	Bengaluru
Fintech	Scalapay	\$1	2/23/2022	Italy	Milan
Finttech	Digital Currency Group	\$10	11/1/2021	United States	New York
Hardware	Gaussian Robotics	\$1.1	11/10/2021	China	Shanghai
Health	Omada Health	\$1	2/23/2022	United States	San Francisco
Hopu Investment Management, Boyu Capital, DC Thomson Ventures	Trax	\$1.3	7/22/2019	Singapore	Artificial intelligence
Internet software & services	Veev	\$1	2/24/2022	United States	San Mateo
Jungle Ventures, Accel, Venture Highway	Moglix	\$2.6	5/17/2021	Singapore	E-commerce & direct-to-consumer
Kuang-Chi	HyalRoute	\$3.5	5/26/2020	Singapore	Mobile & telecommunications
Mobile & telecommunications	PicsArt	\$1	8/26/2021	United States	San Francisco
Mundi Ventures, Doqling Capital Partners, Activant Capital	bolttech	\$1	7/1/2021	Singapore	Fintech

	Company	Valuation (\$B)	Date Joined	Country	City
Industry					
Other	InFarm	\$1	12/16/2021	Germany	Berlin
Sequoia Capital China, ING, Alibaba Entrepreneurs Fund	WeLab	\$1	11/8/2017	Hong Kong	Fintech
Sequoia Capital China, Shunwei Capital Partners, Qualgro	PatSnap	\$1	3/16/2021	Singapore	Internet software & services
Sequoia Capital, Thoma Bravo, Softbank	FTX	\$32	7/20/2021	Bahamas	Fintech
SingTel Innov8, Alpha JWC Ventures, Golden Gate Ventures	Carro	\$1	6/14/2021	Singapore	E-commerce & direct-to-consumer
Supply chain, logistics, & delivery	Scandit	\$1	2/9/2022	Switzerland	Zurich
Temasek, Guggenheim Investments, Qatar Investment Authority	ONE	\$1	12/8/2021	Singapore	Internet software & services
Tiger Global Management, Tiger Brokers, DCM Ventures	Amber Group	\$3	6/21/2021	Hong Kong	Fintech
Travel	VOI	\$1	12/21/2021	Sweden	Stockholm
Vertex Ventures SE Asia, Global Founders Capital, Visa Ventures	NIUM	\$1	7/13/2021	Singapore	Fintech
Vision Plus Capital, GSR Ventures, ZhenFund	Advance Intelligence Group	\$2	9/23/2021	Singapore	Artificial intelligence

```
In [ ]: # Displaying the dictionary of industry groups with their corresponding row indi  
sectore.groups
```

```

Out[ ]: {'500 Global, Rakuten Ventures, Golden Gate Ventures': [757], 'Andreessen Horowitz, DST Global, IDG Capital': [951], 'Artificial Intelligence': [333, 550, 656, 697, 991, 999, 1011], 'Artificial intelligence': [0, 27, 80, 81, 82, 88, 95, 106, 117, 130, 141, 148, 171, 177, 216, 236, 294, 304, 313, 353, 372, 375, 376, 385, 387, 390, 396, 398, 415, 429, 437, 447, 472, 481, 489, 504, 513, 573, 595, 622, 636, 690, 692, 693, 715, 722, 732, 754, 783, 794, 803, 830, 838, 840, 847, 876, 885, 892, 895, 898, 903, 904, 917, 931, 940, 942, 944, 948, 977, 1018, 1024], 'Auto & transportation': [34, 62, 63, 94, 126, 131, 140, 160, 233, 237, 266, 287, 310, 322, 339, 392, 465, 474, 505, 530, 580, 588, 600, 680, 727, 728, 784, 828, 863], 'B Capital Group, Monk's Hill Ventures, Dynamic Parcel Distribution': [963], 'Consumer & retail': [21, 29, 39, 108, 179, 361, 389, 399, 412, 479, 480, 520, 535, 672, 681, 721, 742, 805, 806, 825, 905, 962, 982, 1009, 1012], 'Cybersecurity': [55, 61, 64, 76, 89, 103, 156, 162, 222, 269, 305, 309, 323, 328, 329, 331, 337, 343, 344, 400, 448, 475, 477, 487, 506, 542, 556, 587, 617, 618, 634, 662, 667, 686, 730, 768, 769, 785, 881, 919, 981, 984, 990, 992, 1001, 1008, 1019, 1030, 1035], 'Data management & analytics': [8, 36, 116, 120, 133, 138, 169, 188, 204, 205, 213, 234, 235, 320, 378, 402, 406, 443, 454, 488, 498, 512, 527, 531, 555, 562, 596, 649, 683, 706, 720, 745, 748, 778, 867, 907, 969, 1000, 1015, 1021, 1025], 'Dragonfly Captial, Qiming Venture Partners, DST Global': [911], 'E-commerce & direct-to-consumer': [11, 14, 19, 20, 24, 56, 60, 67, 75, 90, 109, 113, 123, 137, 154, 163, 165, 170, 174, 190, 198, 203, 206, 212, 221, 242, 245, 252, 257, 261, 281, 297, 300, 314, 325, 326, 350, 360, 365, 380, 383, 391, 393, 407, 435, 455, 461, 476, 486, 491, 500, 508, 526, 529, 534, 557, 567, 571, 578, 579, 584, 597, 603, 607, 620, 629, 630, 632, 642, 687, 704, 708, 709, 712, 713, 717, 739, 740, 746, 762, 766, 776, 777, 786, 787, 791, 792, 797, 799, 807, 809, 810, 831, 853, 855, 859, 861, 869, 888, 890, ...], 'Edtech': [13, 17, 155, 201, 214, 225, 230, 247, 264, 268, 275, 283, 307, 457, 532, 613, 688, 702, 723, 767, 800, 801, 819, 845, 857, 886, 920, 964], 'Fintech': [2, 3, 6, 9, 12, 22, 23, 28, 37, 38, 43, 52, 59, 65, 69, 70, 73, 77, 78, 79, 84, 91, 92, 97, 102, 104, 105, 118, 121, 124, 127, 128, 129, 134, 135, 139, 145, 147, 157, 166, 172, 175, 178, 181, 189, 202, 219, 224, 228, 231, 232, 243, 246, 248, 249, 254, 256, 262, 263, 271, 284, 285, 288, 289, 291, 293, 295, 298, 299, 311, 327, 335, 338, 341, 342, 345, 348, 356, 359, 366, 367, 377, 379, 384, 388, 408, 411, 414, 418, 419, 421, 423, 424, 427, 430, 432, 439, 444, 446, 458, ...], 'Finttech': [48], 'Hardware': [18, 30, 35, 101, 146, 153, 158, 192, 229, 253, 340, 351, 397, 440, 453, 473, 503, 522, 545, 546, 581, 673, 780, 817, 820, 833, 834, 836, 875, 915, 950, 953, 986], 'Health': [26, 31, 66, 72, 85, 87, 98, 111, 125, 144, 149, 152, 180, 196, 197, 207, 215, 240, 250, 273, 276, 334, 369, 409, 413, 416, 436, 451, 463, 469, 482, 483, 499, 517, 540, 565, 574, 590, 591, 624, 635, 637, 638, 641, 674, 699, 711, 725, 733, 735, 750, 782, 802, 808, 816, 821, 854, 858, 883, 893, 897, 913, 914, 946, 955, 965, 974, 1007, 1034], 'Hopu Investment Management, Boyu Capital, DC Thomson Ventures': [639], 'Internet software & services': [5, 16, 25, 32, 33, 42, 44, 45, 46, 49, 50, 53, 68, 74, 86, 93, 96, 100, 110, 112, 115, 122, 143, 150, 151, 159, 161, 167, 168, 182, 183, 184, 186, 187, 191, 195, 200, 209, 210, 211, 218, 220, 223, 226, 227, 251, 258, 270, 272, 274, 277, 278, 279, 280, 282, 286, 290, 296, 302, 312, 315, 316, 317, 319, 324, 330, 332, 336, 349, 358, 362, 363, 368, 370, 381, 382, 386, 394, 403, 417, 420, 422, 426, 433, 442, 445, 449, 452, 456, 459, 460, 467, 468, 471, 484, 492, 496, 501, 507, 509, ...], 'Jungle Ventures, Accel, Venture Highway': [318], 'Kuang-Chi': [217], 'Mobile & telecommunications': [57, 107, 136, 176, 185, 194, 199, 239, 259, 301, 308, 373, 404, 405, 434, 478, 525, 536, 599, 614, 619, 644, 657, 676, 738, 779, 788, 789, 795, 811, 815, 818, 823, 826, 842, 860, 949], 'Mundi Ventures, Doqling Capital Partners, Activant Capital': [928], 'Other': [1, 4, 47, 51, 54, 99, 132, 164, 193, 208, 238, 244, 267, 321, 346, 352, 355, 357, 364, 374, 395, 428, 431, 470, 497, 523, 541, 566, 570, 609, 627, 633, 640, 643, 659, 675, 677, 691, 694, 729, 741, 773, 790, 813, 839, 844, 856, 862, 865, 874, 877, 901, 910, 987, 1003, 1013], 'Sequoia Capital China, ING, Alibaba Entrepreneurs Fund': [814], 'Sequoia Capital China, Shunwei Capital Partners, Qualgro': [882], 'Sequoia Capital, Thoma Bravo, Softbank': [10], 'SingTel Innov8, Alpha JWC Ventures, Golden Gate Ventures': [916], 'Supply chain, logistics,

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& delivery': [7, 15, 40, 41, 71, 114, 119, 173, 241, 255, 265, 303, 306, 347, 354, 371, 410, 425, 450, 518, 547, 575, 576, 582, 601, 648, 655, 670, 679, 696, 698, 718, 724, 736, 747, 771, 772, 775, 812, 824, 827, 835, 837, 864, 873, 879, 912, 918, 925, 926, 927, 938, 961, 975, 976, 1016, 1026], 'Temasek, Guggenheim Investments, Qatar Investment Authority': [1014], 'Tiger Global Management, Tiger Brokers, DCM Ventures': [292], 'Travel': [58, 83, 142, 260, 401, 441, 528, 533, 623, 651, 743, 832, 846, 1010], 'Vertex Ventures SE Asia, Global Founders Capital, Visa Ventures': [939], 'Vision Plus Capital, GSR Ventures, ZhenFund': [438]}
```

```
In [ ]: # Accessing the 82nd row (index 81) of the DataFrame to view all column values f
company.iloc[81, :]
```

```
Out[ ]:
```

	81
Company	Argo AI
Valuation (\$B)	\$7.25
Date Joined	7/12/2019
Country	United States
City	Pittsburgh
Industry	Artificial intelligence
Select Inverstors	Volkswagen Group, Ford Autonomous Vehicles
Founded Year	NaN
Total Raised	\$500M
Financial Stage	NaN
Investors Count	2.0
Deal Terms	1.0
Portfolio Exits	NaN

dtype: object

```
In [ ]: # Retrieving all companies that belong to the 'Artificial intelligence' industry
sectore.get_group('Artificial intelligence')
```


Out[]:

	Company	Valuation (\$B)	Date Joined	Country	City	Industry	Select Investors
0	Bytedance	\$140	4/7/2017	China	Beijing	Artificial intelligence	Sequoia Capital China, SIG Asia Investments, S...
27	Faire	\$12.4	10/30/2019	United States	San Francisco	Artificial intelligence	Khosla Ventures, Forerunner Ventures, Sequoia ...
80	Scale AI	\$7.3	8/5/2019	United States	San Francisco	Artificial intelligence	Accel, Y Combinator, Index Ventures
81	Argo AI	\$7.25	7/12/2019	United States	Pittsburgh	Artificial intelligence	Volkswagen Group, Ford Autonomous Vehicles
82	Gong	\$7.25	8/12/2020	United States	Palo Alto	Artificial intelligence	Norwest Venture Partners, Next World Capital, ...
...	...	...	...	...	...	...	...
944	Snorkel AI	\$1	8/9/2021	United States	Palo Alto	Artificial intelligence	Greylock Partners, Google Ventures, BlackRock
948	Shield AI	\$1	8/24/2021	United States	San Diego	Artificial intelligence	Andreessen Horowitz, Homebrew, Point72 Ventures
977	Augury	\$1	10/26/2021	United States	New York	Artificial intelligence	Lerer Hippeau, Munich Re Ventures, Eclipse Ven...
1018	Placer.ai	\$1	1/12/2022	United States	Los Altos	Artificial intelligence	Fifth Wall Ventures, JBV Capital, Array Ventures
1024	Betterfly	\$1	2/1/2022	Chile	Santiago	Artificial intelligence	QED Investors, DST Global, Endeavor

71 rows × 13 columns

```
In [ ]: # Getting the number of rows and columns for the 'Artificial intelligence' indus  
sectore.get_group('Artificial intelligence').shape
```

```
Out[ ]: (71, 13)
```

```
In [ ]: # Calculating the sum of all numeric columns for each industry group  
sectore.sum()
```

Out[]:

		Company	Valuation (\$B)	
Industry				
500 Global, Rakuten Ventures, Golden Gate Ventures		Carousell	\$1.1	
Andreessen Horowitz, DST Global, IDG Capital		Cider	\$1	
Artificial Intelligence	UniphoreNotCoTUNGEEsSmartMoreOwkinAnyscaleHaoma...		2.5151.3	2/
			1.211\$1	
Artificial intelligence	BytedanceFaireScale AIArgo AIGongAutomation An...		14012.4	4,
			7.37.25	
			7.256.8	
			6.35.25.3	
			554....	
Auto & transportation	ZongMu TechnologyNuroBoltOla CabsHello TransTe...		11.48.6	6,
			8.47.5555	
			4.43.35	
			3.313	
			3.3\$3....	
B Capital Group, Monk's Hill Ventures, Dynamic Parcel Distribution		Ninja Van	\$1	
Consumer & retail	Yuanqi SenlinJUUL LabsWeilongEasyhomelImpossible...		1512	3/
			10.885.78	
			42.25222	
			1.71.7	
			1.59\$1....	
Cybersecurity	TaniumSnykLaceworkNetskope1PasswordWizSocureAr...		98.68.3	3,
			7.56.86	
			4.54.33.5	
			32.75	
			2.74\$2...	
Data management & analytics	DatabricksCelonisFiveTranCollibraSambaNova Sys...		38115.6	2,
			5.25554.2	
			43.73.7	
			3.6	
			3.35\$3.3...	
Dragonfly Captial, Qiming Venture Partners, DST Global		Matrixport	\$1.05	
E-commerce & direct-to-consumer	FanaticsXiaohongshuSHEINgoPuffOpenSeaChehaoduo...		27201515	6,
			13.398.7	
			87.56.6	

		Company	Valuation (\$B)
Industry			
			5.77
			5.7\$5.1...
Edtech	BYJU'sYuanfudaoVIPKidArticulateCourse HeroGoSt...		2115.54.5
			3.753.6
			3.53.44
			3.2333
			3\$2.7...
Fintech	StripeKlarnaCheckout.comRevolutChimeRipplePlai...		9545.640
			332515
			13.412.3
			1110.210
		9.23...	
Finttech	Digital Currency Group		\$10
Hardware	DJI InnovationsBitmain TechnologiesGlobal Swit...		151211.1
			64.734.58
			4.5843.5
			3.152.4
			2....
Health	Devoted HealthBiosplice TherapeuticsTempusCari...		12.612
			8.17.8377
			6.25.75
			4.85.85
			4.6\$4...
Hopu Investment Management, Boyu Capital, DC Thomson Ventures		Trax	\$1.3
Internet software & services	CanvaMiroGrammarlyGoodLeapAirtableFigmaRedditT...		4017.513
			1211.710
			101010
			9.59.59.2
			8...
Jungle Ventures, Accel, Venture Highway		Moglix	\$2.6
Kuang-Chi	HyalRoute		\$3.5
Mobile & telecommunications	NianticAttentiveMoon ActiveYello MobileClubhou...		95.99544
			43.83.33
			2.82.75
			2.19222...
Mundi Ventures, Doqling Capital Partners, Activant Capital		bolttech	\$1

		Company	Valuation (\$B)
Industry			
Other	SpaceX Epic Games Thrasio HEYTEA Northvolt Black Un...		100.342
			109.28
			9.086.15 12
			4.243.7
			3.33.2...
Sequoia Capital China, ING, Alibaba Entrepreneurs Fund		WeLab	\$1
Sequoia Capital China, Shunwei Capital Partners, Qualgro		PatSnap	\$1
Sequoia Capital, Thoma Bravo, Softbank		FTX	\$32
SingTel Innov8, Alpha JWC Ventures, Golden Gate Ventures		Carro	\$1
Supply chain, logistics, & delivery	Instacart J&T Express Swiggy Lalamove Flexport RELE...		392010.7
			1085.7
			5.254.1 12
			3.33.13
			2.8\$2....
Temasek, Guggenheim Investments, Qatar Investment Authority		ONE	\$1
Tiger Global Management, Tiger Brokers, DCM Ventures		Amber Group	\$3
Travel	OYO Rooms TripActions Hopper Traveloka Mafengwo Tie...		9.67.2553
			221.51.5
			1.351.3 9/
			1.1111
Vertex Ventures SE Asia, Global Founders Capital, Visa Ventures		NIUM	\$1
Vision Plus Capital, GSR Ventures, ZhenFund		Advance Intelligence Group	\$2

```
In [ ]: # Reading the 'deliveries.csv' file into a DataFrame named 'delivery'
delivery = pd.read_csv('/content/deliveries.csv')

# Reading the 'matches.csv' file into a DataFrame named 'match'
match = pd.read_csv('/content/matches.csv')
```

```
In [ ]: # Displaying the first 5 rows of the 'delivery' DataFrame

delivery.head()
```

```
Out[ ]: match_id  inning  batting_team  bowling_team  over  ball  batsman  non_striker  bo
```

	match_id	inning	batting_team	bowling_team	over	ball	batsman	non_striker	bo
0	1	1	Sunrisers Hyderabad	Royal Challengers Bangalore	1	1	DA Warner	S Dhawan	
1	1	1	Sunrisers Hyderabad	Royal Challengers Bangalore	1	2	DA Warner	S Dhawan	
2	1	1	Sunrisers Hyderabad	Royal Challengers Bangalore	1	3	DA Warner	S Dhawan	
3	1	1	Sunrisers Hyderabad	Royal Challengers Bangalore	1	4	DA Warner	S Dhawan	
4	1	1	Sunrisers Hyderabad	Royal Challengers Bangalore	1	5	DA Warner	S Dhawan	

5 rows × 21 columns



```
In [ ]: # Grouping the 'delivery' DataFrame by the 'batsman' column and storing the resu
run = delivery.groupby('batsman')
```

```
In [ ]: # Retrieving all delivery records for the batsman 'V Kohli'

run.get_group('V Kohli')
```

Out[]:

	match_id	inning	batting_team	bowling_team	over	ball	batsman	non_striker
2590	12	1	Royal Challengers Bangalore	Mumbai Indians	1	2	V Kohli	CH Gay
2591	12	1	Royal Challengers Bangalore	Mumbai Indians	1	3	V Kohli	CH Gay
2593	12	1	Royal Challengers Bangalore	Mumbai Indians	1	5	V Kohli	CH Gay
2594	12	1	Royal Challengers Bangalore	Mumbai Indians	1	6	V Kohli	CH Gay
2597	12	1	Royal Challengers Bangalore	Mumbai Indians	2	1	V Kohli	CH Gay
...	...	...	...	...	...	...	...	...
150410	636	2	Royal Challengers Bangalore	Sunrisers Hyderabad	12	5	V Kohli	AB c Villie
150413	636	2	Royal Challengers Bangalore	Sunrisers Hyderabad	13	2	V Kohli	AB c Villie
150414	636	2	Royal Challengers Bangalore	Sunrisers Hyderabad	13	3	V Kohli	AB c Villie
150415	636	2	Royal Challengers Bangalore	Sunrisers Hyderabad	13	4	V Kohli	AB c Villie
150416	636	2	Royal Challengers Bangalore	Sunrisers Hyderabad	13	5	V Kohli	AB c Villie

3494 rows × 21 columns



In []:

```
# Calculating the total runs scored by each batsman,
# sorting them in descending order,
# and displaying the top 10 highest run-scorers

run['batsman_runs'].sum().sort_values(ascending=False).head(10)
```

Out[]: **batsman_runs**

batsman	
SK Raina	4548
V Kohli	4423
RG Sharma	4207
G Gambhir	4132
DA Warner	4014
RV Uthappa	3778
CH Gayle	3651
S Dhawan	3561
MS Dhoni	3560
AB de Villiers	3486

dtype: int64

```
In [ ]: # Creating a mask to filter rows where the batsman scored exactly 6 runs (i.e.,
mask = delivery['batsman_runs'] == 6

# Applying the mask to create a new DataFrame containing only sixes
new_delivery = delivery[mask]
```

```
In [ ]: # Getting the total number of sixes hit (i.e., number of rows in new_delivery)

new_delivery.shape[0]
```

Out[]: 6523

```
In [ ]: # Counting the number of sixes hit by each batsman,
# sorting them in descending order,
# and displaying the top 5 batsmen with the most sixes

new_delivery.groupby('batsman')['batsman_runs'].count().sort_values(ascending=False)
```

Out[]: **batsman_runs**

batsman	
CH Gayle	266
SK Raina	174
RG Sharma	173
DA Warner	160
V Kohli	160

dtype: int64


```
In [ ]: # Creating a new DataFrame 'vk' that contains all delivery records where the batsman is V Kohli
vk = delivery[delivery['batsman'] == 'V Kohli']

In [ ]: # Calculating the total runs scored by V Kohli against each bowling team,
# sorting the teams in descending order of runs conceded,
# and displaying the top 5 teams he scored the most against
vk.groupby('bowling_team')['batsman_runs'].sum().sort_values(ascending=False).head(5)
```

Out[]: **batsman_runs**

bowling_team	
Chennai Super Kings	706
Delhi Daredevils	661
Kings XI Punjab	483
Mumbai Indians	447
Sunrisers Hyderabad	439

dtype: int64

```
In [ ]: # Defining a function to calculate the top 3 teams against which a given batsman has scored the most runs
def run_scored(batsman_name):

    # Filtering the deliveries DataFrame for the specified batsman
    vk = delivery[delivery['batsman'] == batsman_name]

    # Grouping by bowling team, summing runs, sorting in descending order, and returning the top 3 teams
    return vk.groupby('bowling_team')['batsman_runs'].sum().sort_values(ascending=False).head(3)

In [ ]: # Calling the run_scored function for 'V Kohli' to get the top 3 teams he has scored the most runs against
run_scored('V Kohli')
```

Out[]: **batsman_runs**

bowling_team	
Chennai Super Kings	706
Delhi Daredevils	661
Kings XI Punjab	483

dtype: int64

```
In [ ]: # Calling the run_scored function for 'MS Dhoni' to get the top 3 teams he has scored the most runs against
run_scored('MS Dhoni')
```

Out[]:

batsman_runs

bowling_team	
Royal Challengers Bangalore	608
Mumbai Indians	581
Kings XI Punjab	453

dtype: int64

The isin() function

```
In [ ]: #isin()
#find the most destructive death over batsman in the history of ipl
#Strike Rate=(Runs/Balls)*100
#Min batsman 200 balls in over 16-20
```

```
In [ ]: # Creating a new DataFrame 'death_over' containing deliveries from the death over

death_over = delivery[delivery['over'] > 15]
```

```
In [ ]: # Counting the number of balls faced by each batsman in death overs
all_batsman = death_over.groupby('batsman')['batsman_runs'].count()

# Creating a mask to filter batsmen who have faced more than 200 balls in death
x = all_batsman > 200

# Displaying only those batsmen who have faced more than 200 balls in death over
all_batsman[x]
```

Out[]:

batsman_runs

batsman	
A Mishra	225
AB de Villiers	570
AD Mathews	289
AM Rahane	268
AR Patel	229
AT Rayudu	425
BJ Hodge	385
DA Miller	360
DA Warner	228
DJ Bravo	409
DJ Hussey	234
DPMD Jayawardene	246
Harbhajan Singh	418
IK Pathan	465
JA Morkel	425
JH Kallis	231
JP Duminy	518
JP Faulkner	294
KA Pollard	838
KD Karthik	463
KM Jadhav	338
LRPL Taylor	204
MK Pandey	224
MK Tiwary	423
MS Dhoni	1224
NV Ojha	304
P Kumar	268
PP Chawla	311
R Vinay Kumar	235
RA Jadeja	576
RG Sharma	748
RV Uthappa	275

batsman_runs	
batsman	
S Badrinath	283
S Dhawan	243
SK Raina	458
SPD Smith	316
SS Tiwary	300
STR Binny	218
V Kohli	546
WP Saha	295
Y Venugopal Rao	260
YK Pathan	584
Yuvraj Singh	516

dtype: int64

```
In [ ]: # Converting the index of filtered batsmen (who faced more than 200 balls in dea
# into a list and storing it in 'batsman_list'
```

```
batsman_list = all_batsman[x].index.tolist()
```

```
In [ ]: # Displaying the list of batsmen who have faced more than 200 balls in death ove
```

```
batsman_list
```

```
Out[ ]: ['A Mishra',
        'AB de Villiers',
        'AD Mathews',
        'AM Rahane',
        'AR Patel',
        'AT Rayudu',
        'BJ Hodge',
        'DA Miller',
        'DA Warner',
        'DJ Bravo',
        'DJ Hussey',
        'DPMD Jayawardene',
        'Harbhajan Singh',
        'IK Pathan',
        'JA Morkel',
        'JH Kallis',
        'JP Duminy',
        'JP Faulkner',
        'KA Pollard',
        'KD Karthik',
        'KM Jadhav',
        'LRPL Taylor',
        'MK Pandey',
        'MK Tiwary',
        'MS Dhoni',
        'NV Ojha',
        'P Kumar',
        'PP Chawla',
        'R Vinay Kumar',
        'RA Jadeja',
        'RG Sharma',
        'RV Uthappa',
        'S Badrinath',
        'S Dhawan',
        'SK Raina',
        'SPD Smith',
        'SS Tiwary',
        'STR Binny',
        'V Kohli',
        'WP Saha',
        'Y Venugopal Rao',
        'YK Pathan',
        'Yuvraj Singh']
```

```
In [ ]: #Runs scored by all these 43 batsman
        #Balls played by these 43 batsman

        final=delivery[delivery['batsman'].isin(batsman_list)]
```

```
In [ ]: # Calculating the total runs scored by each of the 43 selected batsmen

        runs = final.groupby('batsman')['batsman_runs'].sum()
```

```
In [ ]: # Counting the total number of balls faced by each of the 43 selected batsmen

        balls = final.groupby('batsman')['batsman_runs'].count()
```

```
In [ ]: # Calculating the strike rate for each batsman using the formula: (total runs /
```

```
sr = (runs / balls) * 100
```

```
In [ ]: # Sorting the strike rates of batsmen in descending order to see who has the hig  
sr.sort_values(ascending=False)
```

Out[]:

batsman_runs

batsman	
AB de Villiers	145.129059
YK Pathan	140.751445
KA Pollard	140.621266
DA Warner	138.318401
DA Miller	137.709251
JA Morkel	136.938202
Harbhajan Singh	135.194585
SK Raina	134.995548
MS Dhoni	132.835821
KM Jadhav	130.555556
JP Faulkner	129.802956
SPD Smith	128.507127
RG Sharma	128.497251
WP Saha	127.727646
RV Uthappa	127.635135
V Kohli	126.588437
Yuvraj Singh	126.390244
STR Binny	123.747981
AT Rayudu	123.014257
KD Karthik	123.008475
AR Patel	122.672065
DJ Bravo	122.286822
JP Duminy	121.970624
BJ Hodge	121.422376
AD Mathews	120.868114
DJ Hussey	120.072661
LRPL Taylor	120.070838
RA Jadeja	118.792867
DPMD Jayawardene	118.791064
S Dhawan	118.502496
AM Rahane	117.486549
MK Pandey	116.938453

batsman_runs	
batsman	
IK Pathan	116.751269
S Badrinath	115.742972
SS Tiwary	114.851485
NV Ojha	114.528024
MK Tiwary	114.127424
Y Venugopal Rao	113.872832
PP Chawla	110.278373
R Vinay Kumar	106.666667
JH Kallis	105.936272
P Kumar	105.263158
A Mishra	89.005236

dtype: float64

The Merge function

```
In [ ]: # Merging the 'delivery' and 'match' DataFrames based on match IDs
# 'match_id' column from 'delivery' is matched with 'id' column from 'match'

delivery.merge(match, left_on='match_id', right_on='id')
```


Out[]:

	match_id	inning	batting_team	bowling_team	over	ball	batsman	non_striker
0	1	1	Sunrisers Hyderabad	Royal Challengers Bangalore	1	1	DA Warner	S Dhaw
1	1	1	Sunrisers Hyderabad	Royal Challengers Bangalore	1	2	DA Warner	S Dhaw
2	1	1	Sunrisers Hyderabad	Royal Challengers Bangalore	1	3	DA Warner	S Dhaw
3	1	1	Sunrisers Hyderabad	Royal Challengers Bangalore	1	4	DA Warner	S Dhaw
4	1	1	Sunrisers Hyderabad	Royal Challengers Bangalore	1	5	DA Warner	S Dhaw
...	...	...	...	...	...	...	...	...
150455	636	2	Royal Challengers Bangalore	Sunrisers Hyderabad	20	2	Sachin Baby	CJ Jorda
150456	636	2	Royal Challengers Bangalore	Sunrisers Hyderabad	20	3	Sachin Baby	CJ Jorda
150457	636	2	Royal Challengers Bangalore	Sunrisers Hyderabad	20	4	Iqbal Abdulla	Sach Bak
150458	636	2	Royal Challengers Bangalore	Sunrisers Hyderabad	20	5	Sachin Baby	Iqb Abdul
150459	636	2	Royal Challengers Bangalore	Sunrisers Hyderabad	20	6	Iqbal Abdulla	Sach Bak

150460 rows × 39 columns



In []:

```
# Storing the merged DataFrame of 'delivery' and 'match' into a new DataFrame na
# This allows access to both ball-by-ball and match-level information in one pla

new = delivery.merge(match, left_on='match_id', right_on='id')
```

In []:

```
# Printing the shape (number of rows and columns) of the 'delivery' DataFrame
print(delivery.shape)
```

```
# Printing the shape (number of rows and columns) of the 'match' DataFrame
print(match.shape)
```

```
(150460, 21)
(636, 18)
```

```
In [ ]: # Grouping the merged DataFrame 'new' by 'season' and 'batsman' to calculate tot
# Sorting the result to get the top scorer per season
# Resetting index to turn grouped columns into regular columns
# Dropping duplicates to keep only the top scorer for each season
# Sorting the final result by season in ascending order

new.groupby(['season', 'batsman'])['batsman_runs'].sum().sort_values(ascending=F
```

```
Out[ ]:
```

	season	batsman	batsman_runs
10	2008	SE Marsh	616
14	2009	ML Hayden	572
9	2010	SR Tendulkar	618
11	2011	CH Gayle	608
2	2012	CH Gayle	733
3	2013	MEK Hussey	733
6	2014	RV Uthappa	660
17	2015	DA Warner	562
0	2016	V Kohli	973
7	2017	DA Warner	641

Pivot Table

```
In [ ]: # Creating a mask to filter rows where the batsman scored 6 runs (i.e., hit a si
mask = delivery['batsman_runs'] == 6

# Applying the mask to create a new DataFrame 'six' containing only the sixes
six = delivery[mask]

# Displaying the first 5 rows of the 'six' DataFrame
six.head()
```

Out[]:

	match_id	inning	batting_team	bowling_team	over	ball	batsman	non_striker
10	1	1	Sunrisers Hyderabad	Royal Challengers Bangalore	2	4	DA Warner	S Dhawan
47	1	1	Sunrisers Hyderabad	Royal Challengers Bangalore	8	4	MC Henriques	S Dhawan
75	1	1	Sunrisers Hyderabad	Royal Challengers Bangalore	13	2	Yuvraj Singh	MC Henriques
89	1	1	Sunrisers Hyderabad	Royal Challengers Bangalore	15	3	Yuvraj Singh	MC Henriques
91	1	1	Sunrisers Hyderabad	Royal Challengers Bangalore	15	5	MC Henriques	Yuvraj Singh

5 rows × 21 columns



In []: *# Displaying the shape of the 'six' DataFrame to see the total number of sixes a*
`six.shape`

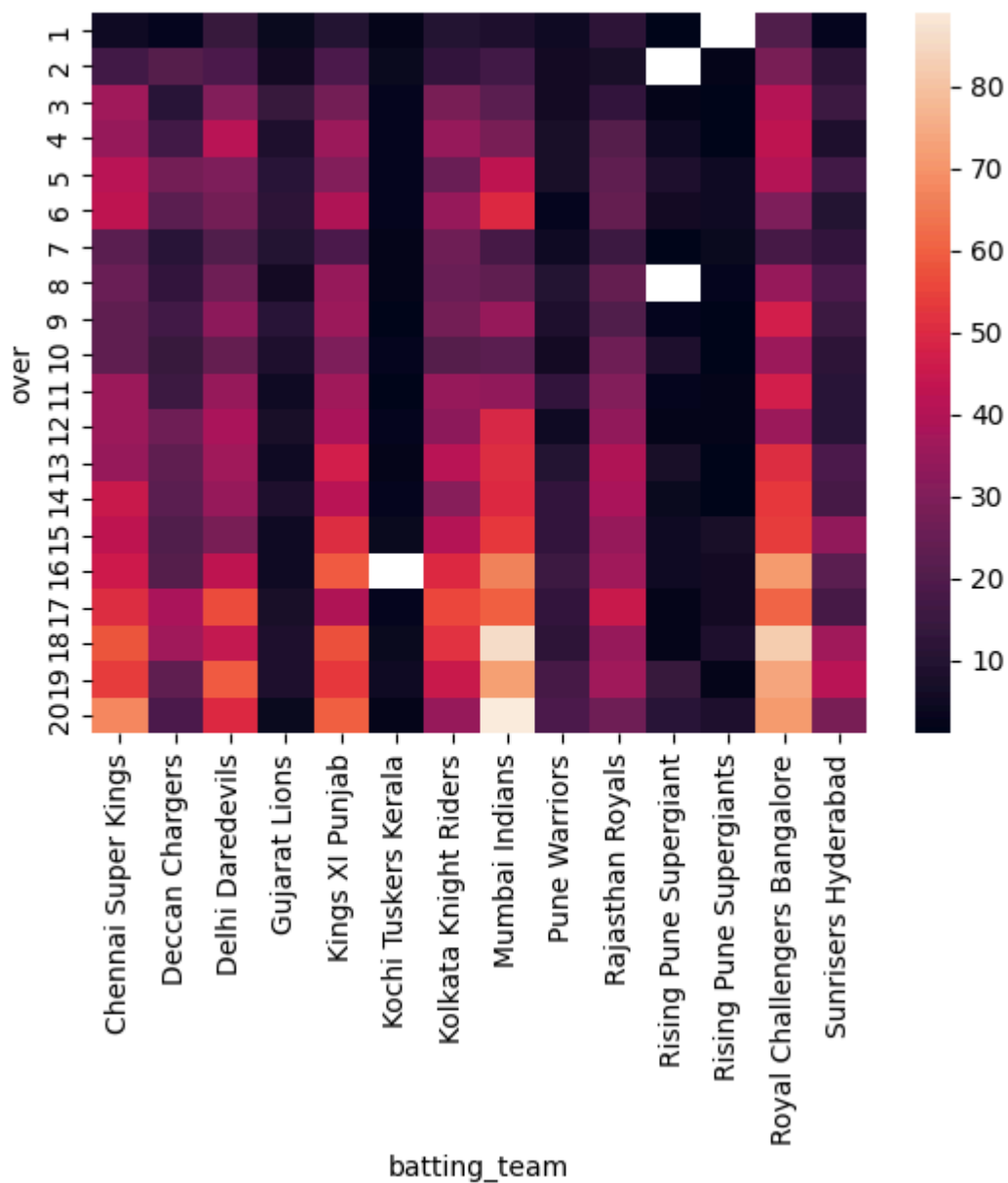
Out[]: (6523, 21)

In []: *# Creating a pivot table to count the number of sixes hit by each batting team i*
Rows represent overs, columns represent batting teams, and values are the coun
`pt = six.pivot_table(index='over', columns='batting_team', values='batsman_runs')`

In []: *# Importing the seaborn library for data visualization, especially for statistic*
`import seaborn as sns`

In []: *# Creating a heatmap to visualize the number of sixes hit by each team in each o*
Darker colors represent higher counts of sixes
`sns.heatmap(pt)`

Out[]: <Axes: xlabel='batting_team', ylabel='over'>



The rename fuction

```
In [ ]: # Renaming the 'city' column in the 'match' DataFrame to 'place'

match.rename(columns={'city': 'place'})
```

Out[]:

	id	season	place	date	team1	team2	toss_winner	toss_decision
0	1	2017	Hyderabad	2017-04-05	Sunrisers Hyderabad	Royal Challengers Bangalore	Royal Challengers Bangalore	field
1	2	2017	Pune	2017-04-06	Mumbai Indians	Rising Pune Supergiant	Rising Pune Supergiant	field
2	3	2017	Rajkot	2017-04-07	Gujarat Lions	Kolkata Knight Riders	Kolkata Knight Riders	field
3	4	2017	Indore	2017-04-08	Rising Pune Supergiant	Kings XI Punjab	Kings XI Punjab	field
4	5	2017	Bangalore	2017-04-08	Royal Challengers Bangalore	Delhi Daredevils	Royal Challengers Bangalore	bat
...	...	...	...	...	...	...	...	...
631	632	2016	Raipur	2016-05-22	Delhi Daredevils	Royal Challengers Bangalore	Royal Challengers Bangalore	field
632	633	2016	Bangalore	2016-05-24	Gujarat Lions	Royal Challengers Bangalore	Royal Challengers Bangalore	field
633	634	2016	Delhi	2016-05-25	Sunrisers Hyderabad	Kolkata Knight Riders	Kolkata Knight Riders	field
634	635	2016	Delhi	2016-05-27	Gujarat Lions	Sunrisers Hyderabad	Sunrisers Hyderabad	field
635	636	2016	Bangalore	2016-05-29	Sunrisers Hyderabad	Royal Challengers Bangalore	Sunrisers Hyderabad	bat

636 rows × 18 columns



set_index() and reset_index()

```
In [ ]: # Displaying the first 5 rows of the 'match' DataFrame
match.head(5)
```

Out[]:

	id	season	city	date	team1	team2	toss_winner	toss_decision	r
0	1	2017	Hyderabad	2017-04-05	Sunrisers Hyderabad	Royal Challengers Bangalore	Royal Challengers Bangalore	field	no
1	2	2017	Pune	2017-04-06	Mumbai Indians	Rising Pune Supergiant	Rising Pune Supergiant	field	no
2	3	2017	Rajkot	2017-04-07	Gujarat Lions	Kolkata Knight Riders	Kolkata Knight Riders	field	no
3	4	2017	Indore	2017-04-08	Rising Pune Supergiant	Kings XI Punjab	Kings XI Punjab	field	no
4	5	2017	Bangalore	2017-04-08	Royal Challengers Bangalore	Delhi Daredevils	Royal Challengers Bangalore	bat	no

In []:

```
# Setting the 'id' column as the index of the 'match' DataFrame
# Note: This does not modify the original DataFrame unless inplace=True is speci

match.set_index("id")
```

Out[]:

	season	city	date	team1	team2	toss_winner	toss_decision	res
id								
1	2017	Hyderabad	2017-04-05	Sunrisers Hyderabad	Royal Challengers Bangalore	Royal Challengers Bangalore	field	nor
2	2017	Pune	2017-04-06	Mumbai Indians	Rising Pune Supergiant	Rising Pune Supergiant	field	nor
3	2017	Rajkot	2017-04-07	Gujarat Lions	Kolkata Knight Riders	Kolkata Knight Riders	field	nor
4	2017	Indore	2017-04-08	Rising Pune Supergiant	Kings XI Punjab	Kings XI Punjab	field	nor
5	2017	Bangalore	2017-04-08	Royal Challengers Bangalore	Delhi Daredevils	Royal Challengers Bangalore	bat	nor
...	...	...	...	...	...	...	...	...
632	2016	Raipur	2016-05-22	Delhi Daredevils	Royal Challengers Bangalore	Royal Challengers Bangalore	field	nor
633	2016	Bangalore	2016-05-24	Gujarat Lions	Royal Challengers Bangalore	Royal Challengers Bangalore	field	nor
634	2016	Delhi	2016-05-25	Sunrisers Hyderabad	Kolkata Knight Riders	Kolkata Knight Riders	field	nor
635	2016	Delhi	2016-05-27	Gujarat Lions	Sunrisers Hyderabad	Sunrisers Hyderabad	field	nor
636	2016	Bangalore	2016-05-29	Sunrisers Hyderabad	Royal Challengers Bangalore	Sunrisers Hyderabad	bat	nor

636 rows × 17 columns



In []:

```
# Resetting the index of the 'match' DataFrame to turn the index into a regular
# This is useful if the index was previously set to a column like 'id'

match.reset_index()
```

Out[]:

	index	id	season	city	date	team1	team2	toss_winner	toss_loser
0	0	1	2017	Hyderabad	2017-04-05	Sunrisers Hyderabad	Royal Challengers Bangalore	Royal Challengers Bangalore	
1	1	2	2017	Pune	2017-04-06	Mumbai Indians	Rising Pune Supergiant	Rising Pune Supergiant	
2	2	3	2017	Rajkot	2017-04-07	Gujarat Lions	Kolkata Knight Riders	Kolkata Knight Riders	
3	3	4	2017	Indore	2017-04-08	Rising Pune Supergiant	Kings XI Punjab	Kings XI Punjab	
4	4	5	2017	Bangalore	2017-04-08	Royal Challengers Bangalore	Delhi Daredevils	Royal Challengers Bangalore	
...	...	...	...	...	...	...	...	...	...
631	631	632	2016	Raipur	2016-05-22	Delhi Daredevils	Royal Challengers Bangalore	Royal Challengers Bangalore	
632	632	633	2016	Bangalore	2016-05-24	Gujarat Lions	Royal Challengers Bangalore	Royal Challengers Bangalore	
633	633	634	2016	Delhi	2016-05-25	Sunrisers Hyderabad	Kolkata Knight Riders	Kolkata Knight Riders	
634	634	635	2016	Delhi	2016-05-27	Gujarat Lions	Sunrisers Hyderabad	Sunrisers Hyderabad	
635	635	636	2016	Bangalore	2016-05-29	Sunrisers Hyderabad	Royal Challengers Bangalore	Sunrisers Hyderabad	

636 rows × 19 columns



Handling missing values

1. dropping missing data using dropna()

```
In [ ]: # Displaying the shape (number of rows and columns) of the 'match' DataFrame
```



```
match.shape
```

Out[]: (636, 18)

```
In [ ]: # Dropping all rows from the 'match' DataFrame that contain any missing (NaN) values
match.dropna()
```

Out[]: **id season city date team1 team2 toss_winner toss_decision result dl_applied**



```
In [ ]: # Displaying the shape of the 'match' DataFrame again to see if any changes occurred
# (Note: match.dropna() in the previous step does not modify the DataFrame unless assigned)
match.shape
```

Out[]: (636, 18)

```
In [ ]: # Dropping all columns from the 'match' DataFrame that contain any missing (NaN) values
# and displaying the shape of the resulting DataFrame
match.dropna(axis=1).shape
```

Out[]: (636, 12)

```
In [ ]: # Dropping only those columns from the 'match' DataFrame where all values are missing
# and displaying the shape of the resulting DataFrame
match.dropna(axis=1, how="all").shape
```

Out[]: (636, 17)

```
In [ ]: # Dropping rows from the 'match' DataFrame where the 'result' column has missing values
# and displaying the shape of the resulting DataFrame
match.dropna(subset=["result"]).shape
```

Out[]: (636, 18)

2. filling missing value using fillna()

```
In [ ]: # Filling all missing (NaN) values in the 'match' DataFrame with 0
# Note: This does not modify the original DataFrame unless assigned or inplace=True
match.fillna(0)
```

Out[]:

	id	season	city	date	team1	team2	toss_winner	toss_decision
0	1	2017	Hyderabad	2017-04-05	Sunrisers Hyderabad	Royal Challengers Bangalore	Royal Challengers Bangalore	field
1	2	2017	Pune	2017-04-06	Mumbai Indians	Rising Pune Supergiant	Rising Pune Supergiant	field
2	3	2017	Rajkot	2017-04-07	Gujarat Lions	Kolkata Knight Riders	Kolkata Knight Riders	field
3	4	2017	Indore	2017-04-08	Rising Pune Supergiant	Kings XI Punjab	Kings XI Punjab	field
4	5	2017	Bangalore	2017-04-08	Royal Challengers Bangalore	Delhi Daredevils	Royal Challengers Bangalore	bat
...	...	...	...	...	...	...	...	...
631	632	2016	Raipur	2016-05-22	Delhi Daredevils	Royal Challengers Bangalore	Royal Challengers Bangalore	field
632	633	2016	Bangalore	2016-05-24	Gujarat Lions	Royal Challengers Bangalore	Royal Challengers Bangalore	field
633	634	2016	Delhi	2016-05-25	Sunrisers Hyderabad	Kolkata Knight Riders	Kolkata Knight Riders	field
634	635	2016	Delhi	2016-05-27	Gujarat Lions	Sunrisers Hyderabad	Sunrisers Hyderabad	field
635	636	2016	Bangalore	2016-05-29	Sunrisers Hyderabad	Royal Challengers Bangalore	Sunrisers Hyderabad	bat

636 rows × 18 columns



In []:

```
# Filling missing values in the 'umpire3' column with the string "Not Specified"
# Note: This does not modify the original DataFrame unless assigned or inplace=True

match["umpire3"].fillna("Not Specified")
```

Out[]:

umpire3

0	Not Specified
1	Not Specified
2	Not Specified
3	Not Specified
4	Not Specified
...	...
631	Not Specified
632	Not Specified
633	Not Specified
634	Not Specified
635	Not Specified

636 rows × 1 columns

dtype: object